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Abstract
Engineering surrogate models, such as neural operators, are increasingly evaluated for predicting aerodynamic flow fields due to their computational efficiency relative to classical solvers. However, a purely data-driven model can achieve low global prediction error while still producing physically inconsistent fields—such as velocity predictions that violate local mass continuity. This study evaluates whether a continuity-based regularization term can reduce local continuity residuals during geometric extrapolation while preserving comparable flow-field prediction accuracy. A controlled comparison was conducted between a baseline 3D Fourier Neural Operator (FNO) and a Physics-Informed FNO (PI-FNO) trained with an explicit discrete divergence penalty. Using a rigid 100/25/50 train-validation-test split on a dataset of internal 3D blade geometries, the results show that the PI-FNO substantially reduced the measured continuity residual across all 50 held-out test geometries (a 98.5% reduction in the mean residual). This improvement was accompanied by a measurable increase in Relative L_2 error from 0.1558 to 0.1594. The experiments indicate a direct trade-off between the selected global prediction-accuracy metric and local physical consistency in this experimental setting.

Keywords: Fourier Neural Operator, Physics-Informed Neural Networks, Aerodynamics, Continuity Equation, Surrogate Modeling.

1. Introduction

Data-driven surrogate models are frequently investigated for partial differential equation (PDE) prediction tasks, offering rapid inference speeds for repeated engineering evaluations. Neural operators, which learn mappings between infinite-dimensional function spaces, have demonstrated strong data-fitting capabilities for fluid dynamics problems.

The investigation began from a practical limitation of purely data-driven surrogates: low prediction error relative to a dataset does not necessarily imply physical consistency. Minimizing a global structural error metric against training data does not inherently satisfy the governing physical equations. Consequently, a surrogate can produce predictions that are numerically close to a reference flow field while exhibiting physically undesirable local artifacts. This vulnerability is especially pronounced during geometric extrapolation, where the model encounters domain shapes absent from the training distribution.

One strategy to mitigate unphysical predictions without abandoning the rapid inference of neural operators is physics-informed regularization. By explicitly penalizing violations of physical laws during training, the network is encouraged to prioritize physically consistent solutions. This investigation evaluates the application of a targeted continuity-based physics regularization to a baseline Fourier Neural Operator applied to aerodynamic flow fields.

2. Problem Statement and Research Objective

In the evaluated baseline model, local continuity violations were observed during geometric extrapolation. Velocity fields produced by the purely data-driven FNO exhibited localized high-frequency residual structures near extrapolated geometry boundaries.

This observation led to a focused engineering question: Can a targeted continuity-based regularization reduce these local continuity residuals during geometric extrapolation while preserving comparable flow-field prediction accuracy?

To examine this question, a continuity-regularized model (PI-FNO) was implemented and evaluated against an identical baseline FNO under a strict, controlled experimental protocol. The investigation did not seek to evaluate whether PI-FNO is universally superior to standard architectures or whether it can replace numerical solvers, but rather to quantify the specific accuracy–consistency trade-off introduced by the regularization.

3. Methodology

3.1 Dataset and Problem Setup

The investigation utilized 3D aerodynamic flow fields and internal blade geometries sourced from the BladeNet dataset. The problem was formulated as predicting a steady-state flow field (density, velocity components, and pressure) given a geometric Signed Distance Field (SDF) and coordinate inputs. All fields were discretely mapped to a unified $128 \times 32 \times 16$ spatial grid.

3.2 Baseline Fourier Neural Operator

The baseline model was a 3D Fourier Neural Operator configured with 8 Fourier modes across all three spatial dimensions and a channel width of 16. The architecture operates by learning transformations involving spectral representations in Fourier space together with local linear transformations, before returning to the spatial domain via an inverse Fourier transform.

3.3 Continuity-Based Physics Regularization

The targeted physical constraint was derived from the steady-state continuity equation for compressible flow, which governs the conservation of mass:

$$\nabla \cdot (\rho \mathbf{u}) = 0$$

where ρ is the fluid density and $\mathbf{u}$ is the velocity vector. Rather than strictly enforcing this continuum equation as a hard constraint, it was implemented as a soft regularization penalty.

3.4 Training Objective

The baseline FNO was trained using a standard Mean Squared Error (MSE) loss, which optimizes for data-fitting accuracy relative to the ground-truth fields.

For the PI-FNO, the training objective incorporated the continuity penalty:

$$ \mathcal{L} = \mathcal{L}

{MSE} + \lambda \mathcal{L}

{Continuity} $$

The term $\mathcal{L}_{Continuity}$ was computed as the mean squared error of the discrete spatial divergence of the density-velocity flux. This divergence was evaluated on the spatial grid using a finite-difference procedure across the internal domain points. The hyperparameter λ controls the relative contribution of the physical constraint.

3.5 Prediction and Normalization Workflow

Both the input states and the target variables were statistically normalized to stabilize training. Because the continuity penalty represents a physical flux, $\mathcal{L}_{Continuity}$ was evaluated on the un-normalized physical fields. During the forward pass, the network's normalized predictions were explicitly un-normalized before calculating the discrete spatial divergence and applying the physical loss.

4. Experimental Design

4.1 Train, Validation, and Test Protocol

The models were evaluated using a strictly separated geometric split: - **100** geometries were used exclusively for training. - **25** geometries were used for validation. - **50** geometries were held out for final testing.

This split ensured that the models were tested entirely on unseen geometric boundary conditions, providing a rigid measure of extrapolation performance.

4.2 Model Configuration

Both the baseline FNO and the PI-FNO utilized identical weight initialization seeds, network architectures, and optimization configurations. The only difference between the models was the inclusion of the continuity penalty in the training objective of the PI-FNO.

4.3 Validation-Based Selection of Physics Weight

The value $\lambda = 1.0$ was selected using the validation set based on the observed balance between continuity improvement and prediction accuracy. The 50 held-out test geometries were strictly isolated and were not used for λ selection or hyperparameter tuning.

4.4 Evaluation Metrics

Performance on the test set was quantified using two primary metrics: 1. **Relative L_2 Error:** A global structural error metric measuring the point-wise deviation between the predicted flow field and the ground truth. 2. **Mean Continuity Residual:** The mean absolute value of the discrete spatial divergence of the predicted field, measuring local physical consistency.

5. Results

5.1 Quantitative Test-Set Comparison

The baseline FNO and the selected PI-FNO ($\lambda = 1.0$) were evaluated on the 50 held-out test geometries. The quantitative outcomes are summarized in Table 1.

Table 1: Final Test-Set Evaluation Metrics (N = 50)

Model	**Relative L_2**	**Mean Continuity Residual**
Baseline FNO	0.1558	1.5468
PI-FNO ($\lambda = 1.0$)	0.1594	0.0225

The integration of the continuity penalty substantially reduced the measured continuity residual by 98.5%. However, this was accompanied by a measured increase of 0.0036 in Relative L_2 error.

5.2 Consistency of Continuity Improvement Across Test Geometries

The reduction in the physical residual was consistent across the dataset. All 50 held-out test geometries exhibited a lower mean continuity residual for the PI-FNO than for the baseline FNO.

5.3 Qualitative Flow-Field and Residual Comparisons

Visual inspection of the continuity residual fields illustrates the effect of the targeted penalty. The PI-FNO suppressed the localized high-frequency residual structures generated by the purely data-driven model.

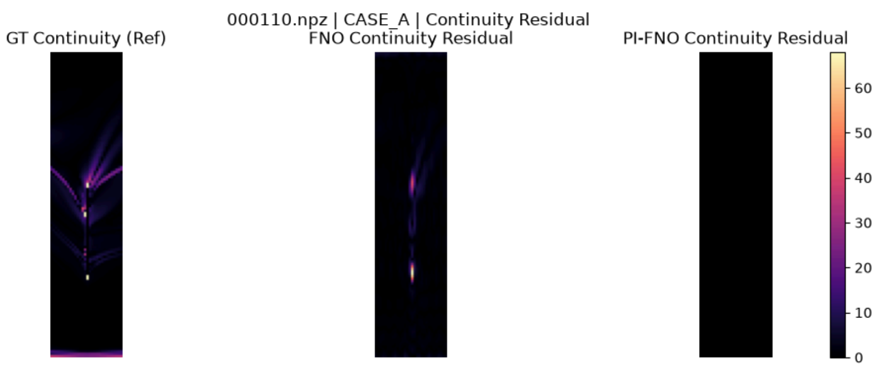

Figure 1: Representative median case comparison. The PI-FNO suppresses localized high-frequency continuity residuals present in the baseline FNO field while retaining macro flow-field structure.

Figure 2: Flow-field prediction across a boundary configuration. Localized continuity residuals observed in the baseline FNO are substantially suppressed in the PI-FNO.

6. Discussion

6.1 The Accuracy–Continuity Trade-off

The results demonstrate a distinct engineering trade-off. While the targeted physical consistency improved by 98.5%, this was achieved at the cost of a measurable degradation (+0.0036) in the global Relative L_2 metric. Figure 3 presents a representative test case selected for a small difference in the global prediction-error metric between the two models. This case illustrates that substantial continuity improvement can occur without a large visible change in the predicted macro flow-field structure. Nonetheless, the quantitative data confirms that the accuracy penalty is a measurable and systemic outcome of the regularization.

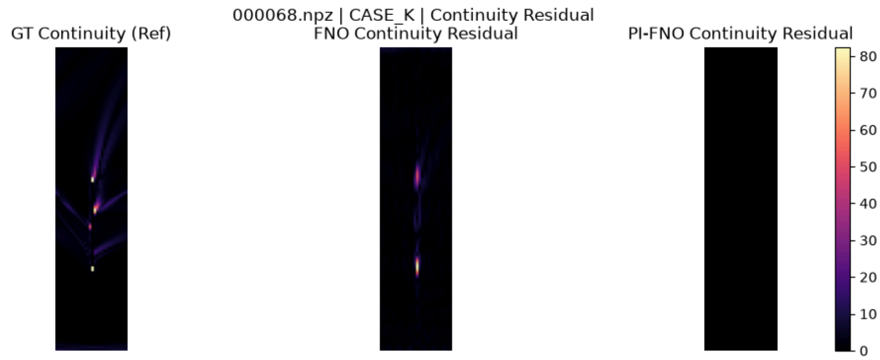

Figure 3: A test case selected for low accuracy difference, illustrating that substantial continuity improvement can occur without a large visible change in the predicted macro flow-field structure.

6.2 Interpretation of the Observed Trade-off

The observed trade-off is consistent with the additional continuity term changing the optimization priorities of the model. By introducing a penalty for the discrete divergence of the fields, the model's objective is shifted such that local continuity consistency is explicitly optimized alongside data fidelity. Therefore, the PI-FNO does not represent a universal accuracy improvement, but rather a measurable exchange between global point-wise prediction accuracy and local physical consistency.

6.3 Implications for Geometric Extrapolation

For the evaluated geometric extrapolation problem, the results suggest that explicitly incorporating a targeted physical residual can reduce a physically undesirable prediction artifact without requiring a change to the core FNO architecture. When engineering workflows prioritize specific local physical metrics alongside general field topology, soft physical constraints offer a tunable mechanism to adjust surrogate behavior.

7. Limitations and Scope

The findings establish behavior within the evaluated geometric extrapolation protocol, rather than universal behavior across all aerodynamic datasets, surrogate architectures, or physical constraints.

1. **Local Constraint vs. Global Conservation:** The implemented continuity residual acts as a local discrete measured constraint and does not guarantee exact global mass conservation across the complete domain.
2. **Accuracy Penalty:** The PI-FNO incurs a measurable Relative L_2 accuracy penalty. The regularization is a trade-off.
3. **Not a CFD Equivalent:** Reducing the local continuity residual does not make the surrogate model equivalent to a converged Computational Fluid Dynamics (CFD) solution.
4. **Specific Regime:** The experimental conclusions are specific to the evaluated BladeNet dataset, geometries, $128 \times 32 \times 16$ grid resolution, baseline FNO architecture, and experimental protocol.
5. **Single Constraint:** Only the continuity constraint was investigated. The behavior of momentum or energy conservation laws under similar regularization was not evaluated.

8. Conclusion

This investigation evaluated whether a continuity-based regularization term could reduce local continuity residuals during geometric extrapolation. The results show that, under the evaluated geometric extrapolation protocol, continuity-based regularization substantially reduced the measured continuity residual across all 50 held-out test geometries. This improvement was accompanied by an increase in Relative L_2 error from 0.1558 to 0.1594. The results therefore demonstrate a measurable trade-off between the selected prediction-accuracy metric and local continuity consistency in this experimental setting.

9. References and Dataset Attribution

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BladeNet

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